Preliminary Design Report

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Section A4
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INTRODUCTION

This preliminary design report aims to detail the process and purpose of analyzing truss structures using computational methods. The project is structured to present real-world engineering practices in structuring through the design, analysis, and optimization of truss structures capable of meeting specific performance criteria. The goal of this analysis is to develop a computer program (MATLAB) that predicts the behavior of truss designs for efficient assessment of their strength, stability, and cost-effectiveness due to varied loads.

The motivation for using a computer program in truss analysis comes from the challenges of manually evaluating such structures. Trusses are composed of multiple interconnected members (simple truss) that form a framework capable of supporting loads. Each member experiences internal forces that must be calculated to ensure the structure's stability. Determining these forces can be labor-intensive and prone to error when solved by hand, especially as the number of members increases. A computational tool facilitates quick and accurate analysis, enabling the fast calculations of various design options with different loads without the time-consuming task of manual calculations.

By developing and using a program for truss analysis, the goal is to deepen the understanding of how internal forces are distributed within a truss design under various loading conditions. This includes recognizing how different design configurations influence the behavior of the truss, particularly in terms of which members experience tension or compression and identifying critical members likely to fail due to buckling.

Through this project, we intend to use our analysis program to evaluate different truss candidates. The goal is to identify designs that meet the specified requirements, such as supporting a minimum load and adhering to cost constraints. The program will also assist in predicting which members are most likely to experience failure due to buckling, thereby informing decisions on material selection and structural changes.

METHODS & ANALYSIS

In order to efficiently evaluate multiple truss designs, we developed a MATLAB function named "trusst_test" (see Appendix for code) to analyze truss structures by processing input matrices that define the truss geometry and loading conditions. **The function accepts six parameters:**

C: Connection matrix, indicating how joints are connected by members.
X: x-coordinates of the joints.
Y: y-coordinates of the joints.
Sx: Reaction force matrix in the x-direction.
Sy: Reaction force matrix in the y-direction.
L: Vector representing the applied external loads.

The function performs several calculations for each design (see "Results" section), including:

1. Estimating the total cost and length of the structure

The code calculates the total length and cost of a truss structure by first determining the lengths of each individual member and then summing these lengths to find the total. For each member, the distance between its connected joints is calculated using the distance formula:

$$length(mem) = \sqrt{(X_{jo2} - X_{jo})^2 + (Y_{jo2} - Y_{jo})^2} \qquad (1)$$

The code then uses the connection matrix "C" to identify which joints are connected by each member. For each member, the Euclidean distance between the two connected joints is computed using their x and y coordinates. These distances are stored in an array "length," and their sum is stored in "tlength," representing the total length of the truss.

Since one inch of truss length costs $1, and one joint costs $10, the cost of the truss is calculated as the sum of two components: the total length of all the members and a fixed cost proportional to the number of joints:

$$cost = tlength + 10 * j \qquad (2)$$

2. Determining the magnitudes and types of forces in the truss members (Compression or tension).

The code calculates the magnitude and type of force in each truss member by solving the system of static equilibrium equations. It first constructs an equilibrium matrix A, which represents the force balance at each joint. For each member, the direction cosines (ratios of x and y components to the member's length) are computed using the coordinates of the connected joints:

$$x_j = X_{jo2} - X_{jo} \ (3) \qquad y_j = Y_{jo2} - Y_{jo} \ (4) \qquad d = \sqrt{x_j^2 + y_j^2} \ (5)$$

$$cosine_x = \frac{x_j}{d} \ (6) \qquad cosine_y = \frac{y_j}{d} \ (7)$$

These values are added to matrix A to account for each member's contribution to the x- and y-direction forces. Reaction forces (Sx and Sy) and external loads (L) are included in A and the right-hand side of the equation. The equation:

$$A * T = L \qquad (8)$$

is then solved for T, where T contains the forces in the members. The magnitude of each force is the absolute value of the corresponding entry in T, and its type is determined by the sign:

- Positive (T > 0): Tension
- Negative (T < 0): Compression

3. Identifying the first member to buckle under load, in addition to its length, predicted buckling strength, and the uncertainty in that strength.

By using the calculated length for each member (see equation 1), the critical buckling force for each member can then be calculated with the formula:

$$P_{crit} = 4863.346 * L^{-2.208} \quad (9)$$

(See Appendix for the graph of Bucking Strength Vs. Length)

For each member, the maximum load weight it can bear is computed by dividing its buckling strength by the force-to-load ratio:

$$W = \frac{-P_{crit}}{R} \qquad (10)$$

Where R is the buckling strength ratio of each member calculated by:

$$R = \frac{T}{total\ load} \quad (11)$$

The member with the smallest W is identified as the first to buckle.

4. Calculating the maximum load weight the structure can bear before failure due to buckling.

By using the smallest determined W of the member that is calculated to buckle first, the maximum load weight of the structure (W_f) can then be determined.

$$W_f = min(W_p) \qquad (13)$$

Here, W_p includes only the positive values of W, representing members under compressive forces (since only compression leads to buckling).

5. Computing the ratio of failure weight to cost, expressed in ounces per dollar (oz/$).

The ratio of failure weight to cost is calculated by the formula:

$$Ratio = \frac{W_f}{cost} \qquad (14)$$

This ratio expresses how much load the structure can bear per dollar spent on construction, providing a measure of cost-efficiency.

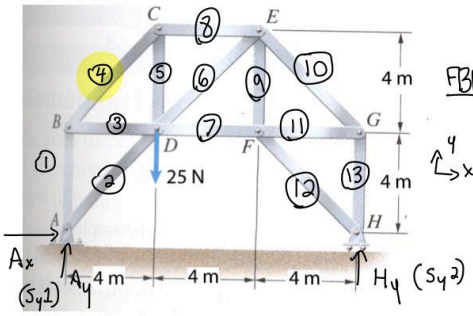
PRACTICE PROBLEM:

Practice Problem Handwritten Solution:

BOSTON UNIVERSITY
College of Engineering
EK 301 Engineering Mechanics I

Truss project computational method validation problem

Determine the loads in each of the members and whether they are in tension or compression. Analyze the loads yourselves using standard equilibrium analysis, and MATLAB (results should match!).



$\tan^{-1}(1) = 45°$

FBD

F_{AB}, A_y

$\Sigma F_y = 0$

$F_{AB} = \boxed{-16.67\,N}$ compression

F_{HG}, H_y

$F_{HG} = \boxed{-8.33\,N}$ compression

$\Sigma F_x = 0$

$A_x = 0$

$\Sigma M_A = 0$

$H_y(12) - 25(4) = \boxed{8.33N}$

$\Sigma F_y = 0$

$A_y + H_y - 25 = 0$

$A_y = \boxed{16.67N}$

ΣFMs

$F_{AD}, F_{FH}, F_{EF} = \boxed{0N}$

FBD2

F_{BC}, F_{BD}, F_{AB}, $-16.67N$

$\Sigma F_y = 0$

$F_{BC}\sin 45 + 16.67 = 0$

$F_{BC} = \boxed{-23.6N}$ compression

$\Sigma F_x = 0$

$F_{BD} - F_{BC}\cos 45 = 0$

$F_{BD} = \boxed{16.67\,N}$ tension

FBD 4

F_{CD}, F_{DE}, F_{BD}, F_{DF}, 25N

$\Sigma F_y = 0$

$16.67 + F_{DE}\sin 45 - 25 = 0$

$F_{DE} = \boxed{11.8\,N}$ tension

$\Sigma F_x = 0$

$F_{DE}\cos 45 + F_{DF} - 16.67 = 0$

$F_{DF} = \boxed{8.33\,N}$ tension

$F_{FG} = \boxed{8.33N}$ tension

FBD 3

F_{BC} $-23.6N$, F_{CD}, F_{CE}

$\Sigma F_x = 0$

$23.6\cos 45 + F_{CE} = 0$

$F_{CE} = \boxed{-16.67N}$ compression

$\Sigma F_y = 0$

$23.6\sin 45 - F_{CD} = 0$

$F_{CD} = \boxed{16.67N}$ tension

FBD 5

$-16.67N$ F_{CE}, E, F_{DE} 11.8N, F_{EG}

$\Sigma F_y = 0$

$F_{DE}\sin 45 + F_{EG}\sin 45 = 0$

11.8N

$F_{EG} = \boxed{-11.8N}$ compression

Practice Problem Code Output (See Appendix for Input code):

```
% EK301, Section A4, Trusst: Esther Xu, Doris Siu, Bryce Hwang 11/12/2024
Load: 25 N
Member forces in N:
m1: 16.667 (C)
m2: 0.000 (C)
m3: 16.667 (T)
m4: 23.570 (C)
m5: 16.667 (T)
m6: 11.785 (T)
m7: 8.333 (T)
m8: 16.667 (C)
m9: 0.000 (T)
m10: 11.785 (C)
m11: 8.333 (T)
m12: 0.000 (C)
m13: 8.333 (C)
Reaction forces in N:
Sx1: 0.00
Sy1: 16.67
Sy2: 8.33
Cost of truss: $140.28
First truss to buckle is member 4
First truss to buckle length: 5.66
Predicted buckling strength: 112.42 N
Total Length: 60.28
Max Load: 112.42 N
Theoretical max load/cost ratio: 0.8013 N/$
```

RESULTS

Design 1

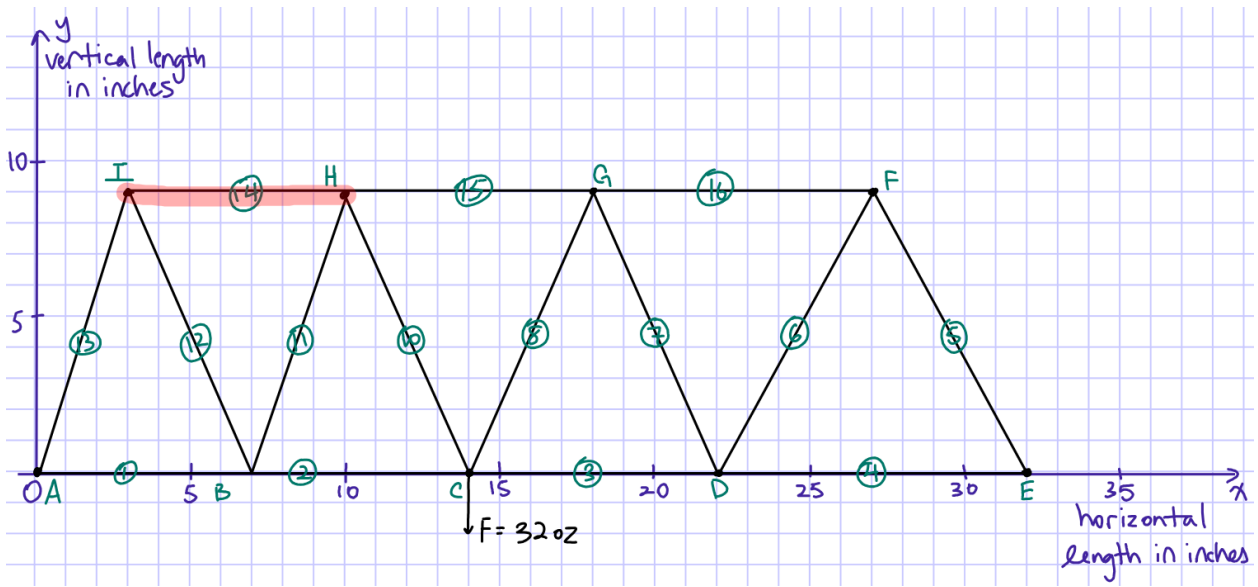


Figure 1: First truss design, with the first member to buckle highlighted in red. Joints are labeled in letters and members are labeled in numbers.

Design 1 Code Output:

```
% EK301, Section A4, Trusst: Esther Xu, Doris Siu, Bryce Hwang 11/12/2024
Load: 32 oz
Member forces in oz:
m1: 6.000 (T)
m2: 20.000 (T)
m3: 21.778 (T)
m4: 7.778 (T)
m5: 16.015 (C)
m6: 16.015 (T)
m7: 15.320 (C)
m8: 15.320 (T)
m9: 19.698 (T)
m10: 18.974 (C)
m11: 19.698 (T)
m12: 18.974 (C)
m13: 14.000 (C)
m14: 28.000 (C)
m15: 15.556 (C)
Reaction forces in oz:
Sx1: -0.00
Sy1: 18.00
Sy2: 14.00
```

```
Cost of truss: $224.96
First truss to buckle is member 14
First truss to buckle length: 8.00
Predicted buckling strength: 56.35 oz
Total Length: 134.96
Max Load: 56.35 oz
Theoretical max load/cost ratio: 0.2505 oz/$
```

Design 1 Data Table:

Member Number	Length (in)	Tension or Compression	Buckling strength and uncertainty (oz)	Magnitude of Force (oz)
1	7	T	-	6
2	7	T	-	20
3	8	T	-	21.778
4	10	T	-	7.778
5	10.296	C	28.22 +-1.54	16.015
6	10.296	T	-	16.015
7	9.849	C	31.15 +-1.54	15.320
8	9.849	T	-	15.320
9	9.849	T	-	19.698
10	9.849	C	31.15 +-1.54	18.974
11	9.849	T	-	19.6977
12	9.487	C	33.82 +-1.54	18.974
13	7	C	66.22 +-1.54	14
14	8	C	49.31 +-1.54	28
15	9	C	38.02 +-1.54	15.56

Table 1: Physical characteristics of all members in Truss Design 1

Design 2

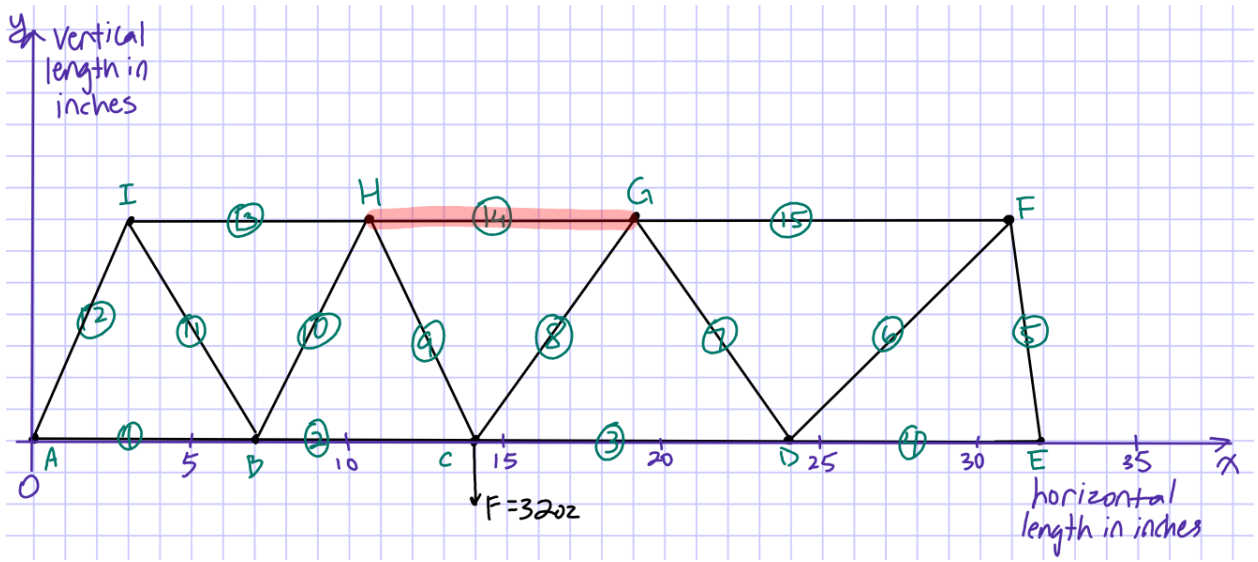


Figure 2: Second truss design, with the first member to buckle highlighted in red. Joints are labeled in letters and members are labeled in numbers.

Design 2 Code Output:

```
% EK301, Section A4, Trusst: Esther Xu, Doris Siu, Bryce Hwang 11/12/2024
Load: 32 oz
Member forces in oz:
m1: 7.7143 (T)
m2: 25.7143 (T)
m3: 26.000 (T)
m4: 2.000 (T)
m5: 14.1421 (C)
m6: 19.799 (T)
m7: 17.2047 (C)
m8: 17.2047 (T)
m9: 20.7315 (T)
m10: 20.7315 (C)
m11: 19.5834 (T)
m12: 19.5834 (C)
m13: 15.4286 (C)
m14: 36.000 (C)
m15: 16.000 (C)
Reaction forces in oz:
Sx1: 0.00
Sy1: 18.00
Sy2: 14.00
Cost of truss: $214.04
First truss to buckle is member 14
First truss to buckle length: 9
```

```
First truss to buckle predicted buckling strength: 38.02 oz
Total Length: 126.61
Max Load: 33.79 oz
Theoretical max load/cost ratio: 0.1579 oz/$
```

Design 2 Data Table:

Member Number	Length (in)	Tension or Compression	Buckling strength and uncertainty (oz)	Magnitude of Force (oz)
1	7	T	-	7.7143
2	7	T	-	25.7142
3	10	T	-	26
4	8	T	-	2
5	7.071	C	64.77 +-1.54	14.1421
6	9.899	T	-	19.799
7	8.602	C	42.03 +-1.54	17.2047
8	8.602	T	-	17.2047
9	8.602	T	-	20.7315
10	8.602	C	42.03 +-1.54	20.7315
11	7.616	T	-	19.5834
12	7.616	C	54.9 +-1.54	19.5834
13	7	C	66.2 +-1.54	15.4286
14	9	C	38.02 +-1.54	36
15	12	C	20.14 +-1.54	16

Table 2: Physical characteristics of all members in Truss Design 2

DISCUSSION and CONCLUSION

Analyzing the results, we found that Design 1 demonstrated a higher maximum load capacity and a better load-to-cost ratio compared to Design 2. This indicates that Design 1 can support more load weight relative to its cost, making it a more robust and cost-efficient option. However, this design comes at a higher overall cost and longer total member length, which could impact constructibility in real-world applications where material cost and accessibility is a major factor.

Design 2, on the other hand, had a lower cost and shorter total length, making it more economical. However, its maximum load capacity was significantly lower, and its load-to-cost ratio was worse compared to Design 1. The first member to buckle in Design 2 also showed a lower buckling strength, which incurs potential stability concerns under higher loads.

The results for both designs align with physical outcomes. The members with longer lengths and compressive forces were more likely to buckle, which is consistent with the principles of truss stability and buckling mechanics. Design 1's higher load capacity but higher cost suggests that it is structurally stronger but less cost-efficient. Design 2's cost demonstrated a more economical design but at the expense of a reduced max load capability.

Going back to the introduction, our primary goal of developing a computational tool to analyze and optimize truss structures was achieved. The analysis highlighted the balance between cost, member length, and the distribution of forces necessary for structural efficiency. The comparison between the designs proved to be valuable for identifying trade-offs and improving the structural design process.

There still are possible factors of uncertainty in our data and sources of error. The calculation of the buckling strength relies on empirical data with a known fit error of ±1.54 oz, which could affect the predicted failure load. Small inaccuracies in floating-point arithmetic within our MATLAB program could introduce minor errors as well. Inaccuracies in joint coordinates or member connections could also mess up the results.

This report taught us the importance of balancing cost and performance when designing truss structures. Computational analysis proved to be an essential tool for quickly assessing multiple designs and understanding force distribution within the structure. We can use our results and knowledge for identifying trade-offs and improving the structural design of many candidates.

Some additional steps next time could also improve our understanding and design. Testing additional designs to achieve higher load-to-cost ratios while maintaining or reducing costs. Adjusting to near-equal member lengths can better distribute forces and prevent premature buckling.

Design 1 is the better candidate due to its higher maximum load capacity and reasonable cost-to-load ratio, making it suitable for scenarios where higher load strength is required. Design 2, while more cost-effective, would require lots of modifications to the members to meet higher load demands. Future iterations could further balance cost and load-bearing performance by adjusting member lengths and joint placements while maintaining peak structural stability.

APPENDIX

a. **Matlab code** used for truss analysis & **Input sample** for Practice Problem

```
% Joint coordinates
X = [0, 0, 4, 4, 8, 8, 12, 12];
Y = [0, 4, 8, 4, 8, 4, 4, 0];

% Connection Matrix
C = [
   1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,0,0;   % Joint A
   1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0,0,0;   % Joint B
   0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0,0,0;   % Joint C
   0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0,0,0;   % Joint D
   0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0,0,0;   % Joint E
   0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1,1,0;   % Joint F
   0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1,0,1;   % Joint G
   0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,1,1;   % Joint H
];

% Reaction force matrices
Sx = [
   1, 0, 0;  % Joint A in x
   0, 0, 0;  % Joint B in x
   0, 0, 0;  % Joint C in x
   0, 0, 0;  % Joint D in x
   0, 0, 0;  % Joint E in x
   0, 0, 0;  % Joint F in x
   0, 0, 0;  % Joint G in x
   0, 0, 0;  % Joint H in x
];
Sy = [
   0, 1, 0;  % Joint A in y
   0, 0, 0;  % Joint B in y
   0, 0, 0;  % Joint C in y
   0, 0, 0;  % Joint D in y
   0, 0, 0;  % Joint E in y
   0, 0, 0;  % Joint F in y
   0, 0, 0;  % Joint G in y
   0, 0, 1;  % Joint H in y
];


% External load vector (L)
L = [0;0;0;0;0;0;0;0; 0;0;0;25;0;0;0;0];

trusst_test(C, X, Y, Sx, Sy, L);

function trusst_test(C, X, Y, Sx, Sy, L)
[j, m] = size(C);
A = zeros(j * 2, m + 3);
A(1:j, (m + 1):(m + 3)) = Sx;
A(j + 1:j * 2, (m + 1):(m + 3)) = Sy; for jo = 1:j
for mem = 1:m
if C(jo, mem) == 1
jo2 = 0;
for jop = 1:j
if C(jop, mem) == 1 && jop ~= jo
jo2 = jop;
break;
```

```
end
end
xj = X(jo2) - X(jo);
yj = Y(jo2) - Y(jo);
d = sqrt(xj^2 + yj^2);
A(jo, mem) = xj / d;
A(jo + j, mem) = yj / d;
end
end
end
T = A\L;
tlength = 0;
length = zeros(1, m);
for mem = 1:m
jc = zeros(1, 2);
i = 1;
for jop = 1:jo
if C(jop, mem) == 1
jc(i) = jop;
i = i + 1;
end
end
xjo = X(jc(1)) - X(jc(2));
yjo = Y(jc(1)) - Y(jc(2)); length(mem) = sqrt((xjo^2) + (yjo^2)); tlength = tlength +
length(mem);
end
maxforce = 4863.346 * length.^(-2.208);
cost = tlength + 10 * j;
load = sum(L);
Tn = (T(1:m))';
R = Tn ./ load;
W = -maxforce ./ R;
Wp = W(W > 0);
Wf = min(Wp);
first2buckle = find(W == Wf, 1);
fprintf('\n%% EK301, Section A4, Trusst: Esther Xu, Doris Siu, Bryce Hwang 11/12/2024\n');
fprintf('Load: %d oz\n', load);
fprintf('Member forces in oz:\n');
for i = 1:m
torc = 'C';
if T(i) > 0
torc = 'T';
end
fprintf('m%d: %.3f (%s)\n', i, abs(T(i)), torc);
end
fprintf('Reaction forces in oz:\n');
fprintf('Sx1: %.2f\n', T(m + 1));
fprintf('Sy1: %.2f\n', T(m + 2));
fprintf('Sy2: %.2f\n', T(m + 3));
fprintf('Cost of truss: $%3.2f\n', cost);
fprintf('First truss to buckle is member %d\n', first2buckle);
fprintf('First truss to buckle length: %3.2f\n', length(first2buckle));
fprintf('Predicted buckling strength: %3.2f oz\n', Wf);
fprintf('Total Length: %3.2f\n', tlength);
fprintf('Max Load: %3.2f oz\n', Wf);
fprintf('Theoretical max load/cost ratio: %.4f oz/$\n', Wf / cost);
end
```

b. Data graph of Buckling Strength versus length

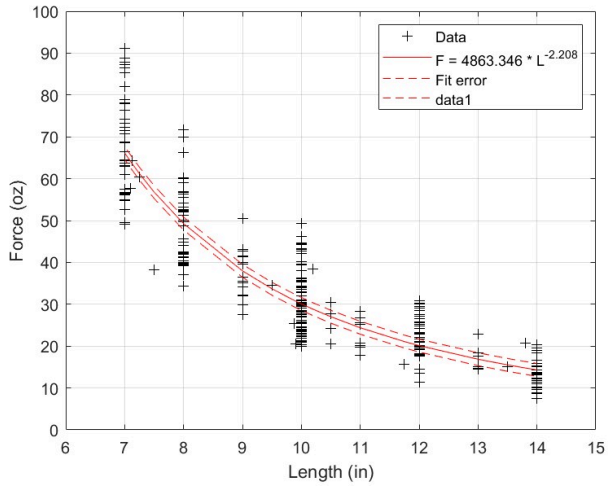


Figure 3: The fit error bar is a constant, resulting in U_fit = +-1.54 oz (95%).